THE UNIVERSITY OF MELBOURNE

comp10002/comp20005
binary search

Support material for Chapter 7 of
Programming, Problem Solving, and Abstraction with C,
by Alistair Moffat

(c) 2023, The University of Melbourne
Prepared by Alistair Moffat, ammoffat@unimelb.edu.au

Searching for: 19

no | maybe | yes!

A[0]	A[1]	A[2]	A[3]	A[4]	A[5]	A[6]	A[7]	A[8]	A[9]	A[10]	A[11]
14	15	17	17	19	19	23	24	25	37	41	52

0 *lo*

hi 12

```
lo, hi ← 0, n
while lo < hi do
    m ← (lo + hi) / 2
    if x < A[m] then
        hi ← m
    else if x > A[m] then
        lo ← m+1
    else
        return m
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0 *lo*

6 *m*

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lo: 0

hi: 6

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0 *lo*   *m* 3   6 *hi*

lo, hi ← 0, *n*
while *lo* < *hi* **do**
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m 3

6 *hi*

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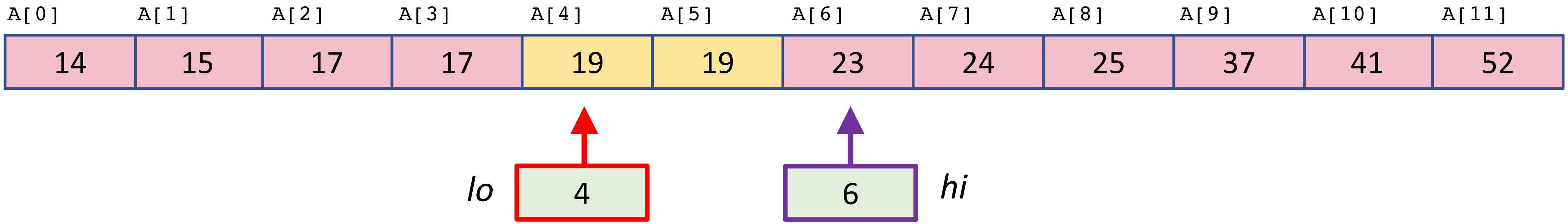


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lo 4

6 *hi*

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lo 4 5 6 *hi*

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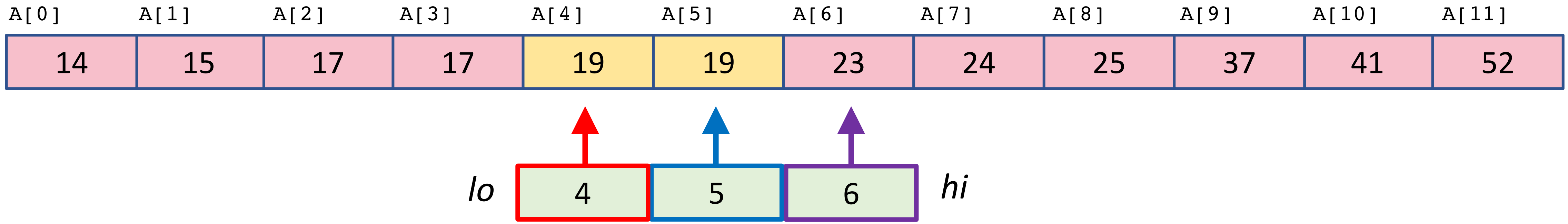


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5

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  else if *x* > *A*[*m*] **then**
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    return *m*
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A[5] == 19

Searching for: 40

no | maybe | yes!

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0 *lo*

6 *m*

hi 12

```
lo, hi ← 0, n
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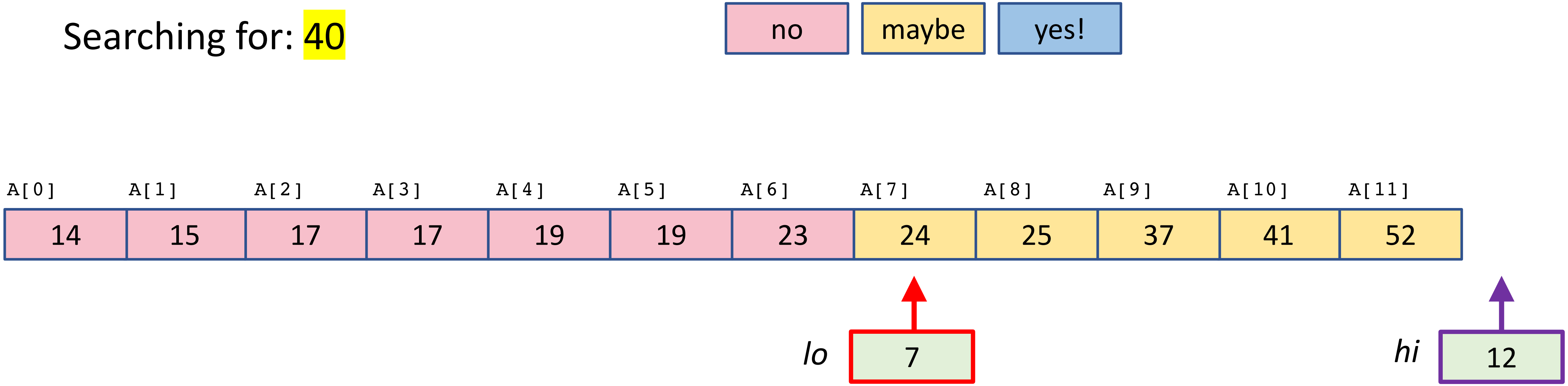

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lo 7

hi 12

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lo 7 *m* 9 *hi* 12

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lo 10 *hi* 12

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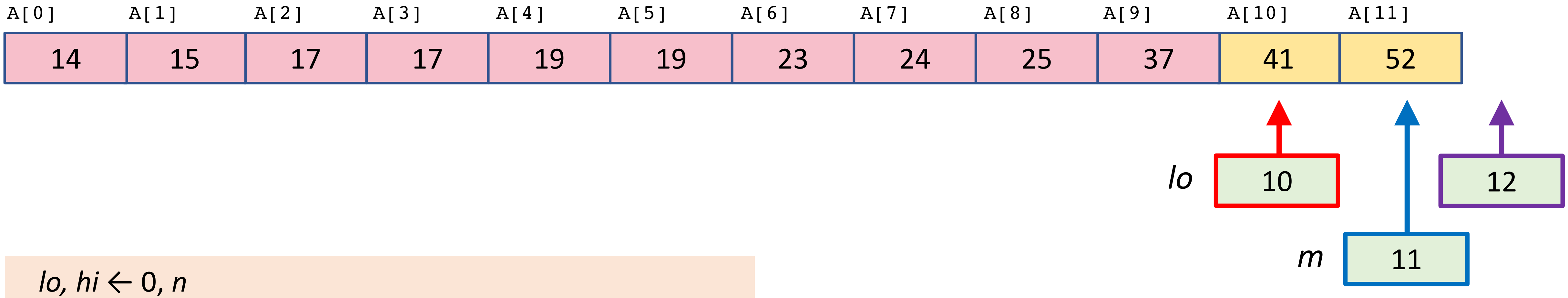


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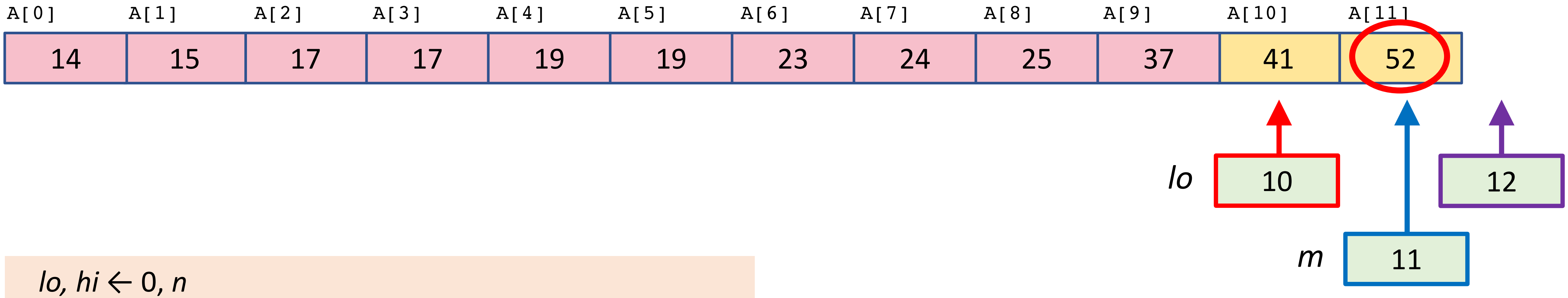


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lo 10 | 11 *hi*

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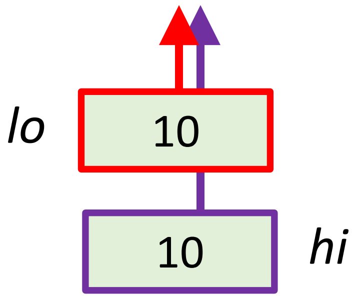


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40 is not in A[0..11]